

TED[15] 5062Reg. No.....

(REVISION-2015)

Signature.....

MODEL QUESTION PAPER

**FIFTH SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOGY  
YARN FORMATION-III**

[Time : 3 hours]

(Maximum marks : 100)

PART - A

(Maximum marks: 10)

Marks

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. List any two modern developments in drafting region of ring-frame.
2. Mention two machines which work on the principle of friction spinning.
3. Define wrapper fibers.
4. State what are the styles of doubling.
5. Define draft constant.

(5x2=10)

PART - B

(Maximum marks: 30)

II. Answer any five of the following questions. Each question carries 6 marks.

1. List the features of energy saving spindle.
2. Write short notes on opposite helically fluted rollers and ABC ring.
3. Explain the principle of twisting in OE rotor spinning with a schematic diagram.
4. Sketch and list the features of S-groove and G-groove rotor.
5. Differentiate S-twist, Z-twist, S on S, and S on Z twist.
6. Explain the procedure of manufacture of tyre-cord.
7. Find out the minimum empty bobbin diameter practical on ring-frame using 42mm diameter ring. Minimum angle of yarn pull is  $23^{\circ}$ .

(5x6=30)

PART - C  
(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT - I

III.a. Explain the offset drafting by INA-V with a neat sketch. (10)

b. Mention the advantages of long length ring frames. (5)

OR

IV. a. Describe the super high draft system with sketch. (10) b.  
Explain the need for compacting elements in ring-frame. (5)

UNIT – II

V.a. Describe the yarn formation in Murata airjet spinning machine.(10)

b. Compare SIRO yarn and ring yarn. (5)

OR

VI.a. Explain the working of DREF-II spinning machine with sketch. (10)

b. List the advantages of OE spinning over ring spinning. (5)

UNIT – III

VII.a. Describe the working of three for one twister with a sketch. (10) b.  
Define reeling and list the objects of reeling. (5)

OR

VIII. a.Explain the working of a fancy doubler with a sketch. (10)b.  
Write short notes on spot yarn, loop yarn. (5)

UNIT – IV

IX.a. Find-out the production/RF/8hrs in kgs from the details given:  
Spindlespeed-2200rpm, T.M-4.0, Hank of roving-2.0, break draft wheel-51T,  
break draft constant-56, main draft-27.27, efficiency-97%,no.of spindles/  
ringframe-1008 (10)

b. Find out the maximum spindle speed practical on ring-frame using 36mm  
diameter ring. Maximum practical traveller speed-45m/sec. (5)

OR

X.a. Find out the production/ ring doubler/ 8hr in kg from data given: Spindle  
speed-12,000rpm, Counts doubled-30s, 36s, 40s, TM-4.0, efficiency-92%, No.of  
spindles/frame-400. (10)

b. Find out the minimum ring diameter practical on ring-frame using 13mm  
diameter empty bobbin. The minimum practical angle of yarn pull is 23°. (5)